

PLUMB

SOFTWARE VERIFICATION & ASSURANCE

PLUMB Flagship Demo

Independent Verification Report

Assessment status	FAILED
Verification coverage	66.7%
Implementation depth	69.5%
Verification ID	63932d95-d6c3-4c9c-85bd-d84a8bdd23b1
Generated	2026-09-15T22:30:57.400456Z

Client-ready assurance statement

This report records the evidence observed during an independent verification run. It distinguishes executed runtime observations, test outcomes, deterministic static findings, measurements, and unsupported capabilities. Requirement-level evidence is authoritative; the summary score is contextual only.

CONFIDENTIAL · Prepared for project stakeholders and client review

Executive decision summary

Decision-oriented summary for engineering leadership, delivery teams, and client stakeholders.

Release recommendation	DO NOT RELEASE / ACCEPT AS COMPLETE
Verified requirements	3 / 7
Failed requirements	4
Inconclusive requirements	0
Tests executed	4 (4 passed)
Security findings	3
Mutation resilience	0 of 1 injected faults detected

Assessment interpretation

The run reached **66.7%** acceptance-criteria verification. 4 requirement(s) are contradicted by recorded evidence and 0 remain unresolved. A passing unit-test suite is not treated as proof of requirement completion.

Priority actions

REQ-002 — Export filtered tasks as CSV

REQ-003 — Merge an incoming batch into an existing task list

REQ-006 — Handle larger duplicate-detection workloads without disproportionate runtime growth.

REQ-007 — Do not expose hardcoded credentials or execute user-controlled commands with unsafe she...

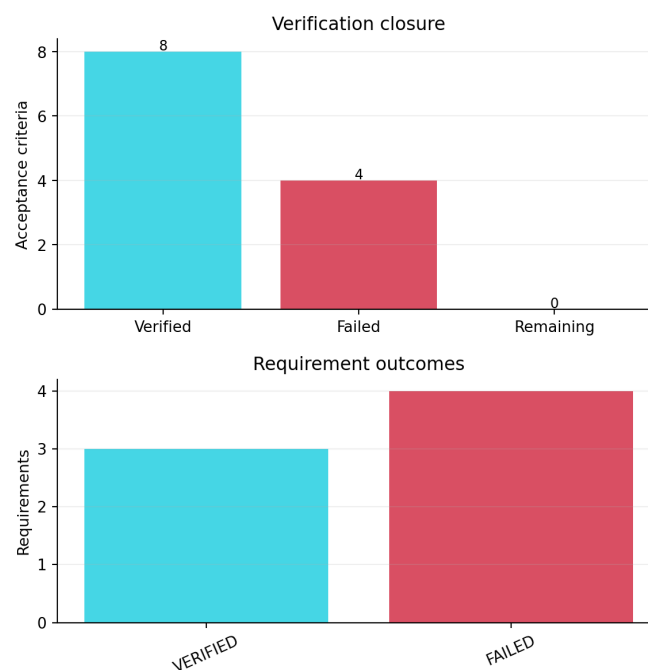
Evidence integrity

Trace ID **63932d95-d6c3-4c9c-85bd-d84a8bdd23b1**. Project hash **46276a962d01b1b695850d87....** All report conclusions are generated from the same stored verification record used by the application.

1. Verification dashboard

A compact view of what reached verification and what remains unresolved.

Metric	Result
Acceptance criteria verified	8 / 12 (66.7%)
Acceptance criteria remaining	4 / 12 (33.3%)
Requirements fully verified	3 / 7 (42.9%)
Implementation depth	69.5%
Verification score	57.9 / 100
Existing test suite	4 passed / 0 failed
Security findings	3
Mutation score	0% (0 of 1 injected faults caught)

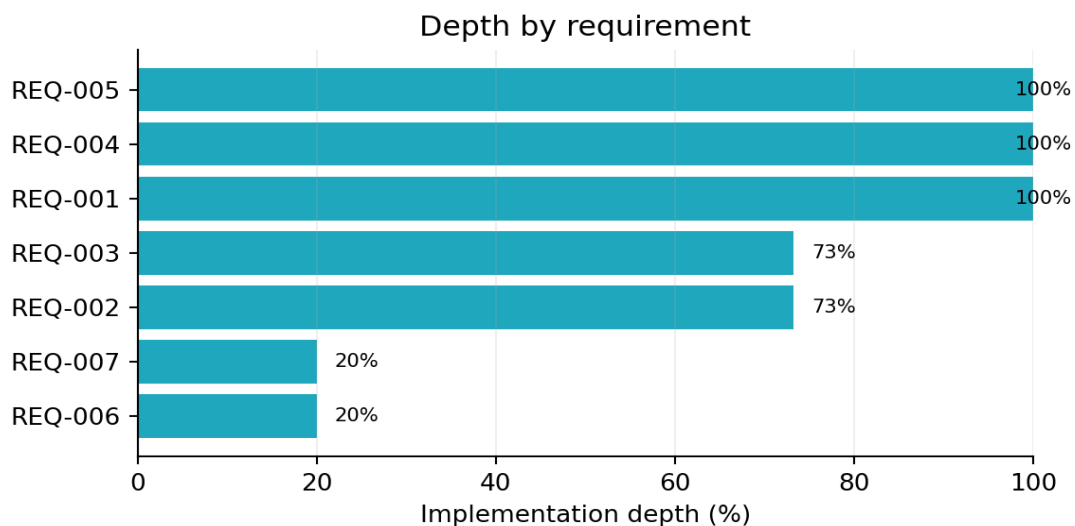


2. Requirements: completion and remaining gap

1. Add tasks with a title, status, and priority; adding a task must preserve the data supplied. 2. Export filtered tasks as CSV; the export must contain only records matching the requested status; an empty filtered result must still return a valid CSV header. 3. Merge an incoming batch into an existing task list; exact duplicate tasks should be removed, but distinct tasks that share a title must remain distinct; insertion order of retained tasks must be preserved. 4. Provide a health endpoint at /health that returns HTTP 200 and a JSON status of ok. 5. Provide an item-count endpoint at /items/count that returns HTTP 200 and count=3 for the demo dataset. 6. Handle larger duplicate-detection workloads without disproportionate runtime growth. 7. Do not expose hardcoded credentials or execute user-controlled commands with unsafe shell semantics.

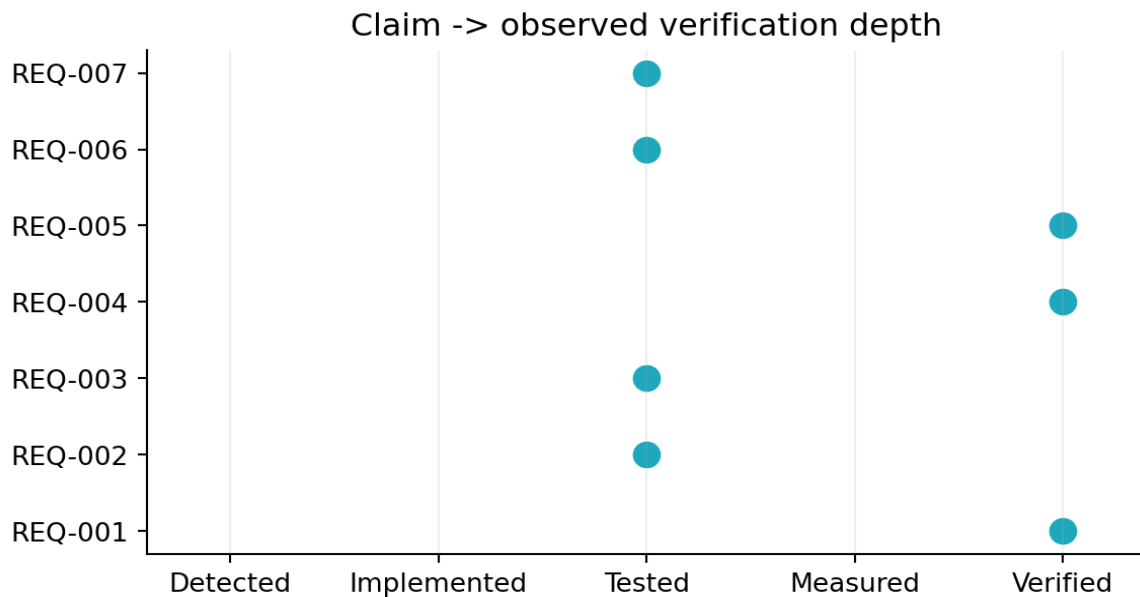
HTTP_CHECK GET /health 200 status=ok HTTP_CHECK GET /items/count 200 count=3

Requirement	Status	Depth	Verified	Remaining
REQ-001 - Add tasks with a title, status, and priority	VERIFIED	100.0%	2/2	0
REQ-002 - Export filtered tasks as CSV	FAILED	73.3%	2/3	1
REQ-003 - Merge an incoming batch into an existing task list	FAILED	73.3%	2/3	1
REQ-004 - Provide a health endpoint at /health that returns HTTP 200 and a JSON status of ok.	VERIFIED	100.0%	1/1	0
REQ-005 - Provide an item-count endpoint at /items/count that returns HTTP 200 and count=3 for th...	VERIFIED	100.0%	1/1	0
REQ-006 - Handle larger duplicate-detection workloads without disproportionate runtime growth.	FAILED	20.0%	0/1	1
REQ-007 - Do not expose hardcoded credentials or execute user-controlled commands with unsafe she...	FAILED	20.0%	0/1	1



Verification depth and evidence posture

This view shows how far each requirement progressed from detection toward direct verification. It is an evidence-stage view, not a completion claim.



How to read this report

Executed runtime observations, test results, static findings, measurements, and unsupported capabilities are kept distinct. FAILED means evidence contradicted the criterion; INCONCLUSIVE means the available evidence was insufficient.

Evidence classes used in this assessment

Evidence class	What it means
Runtime observation	The submitted application was started and a defined behavior was exercised.
Test execution	The submitted automated test suite was executed and its outcome recorded.
Static finding	Source code was inspected deterministically for a specific capability or risk pattern.
Measurement	A bounded timing or scaling probe was executed against submitted code.
Unsupported	The current execution environment cannot safely perform the requested verification.

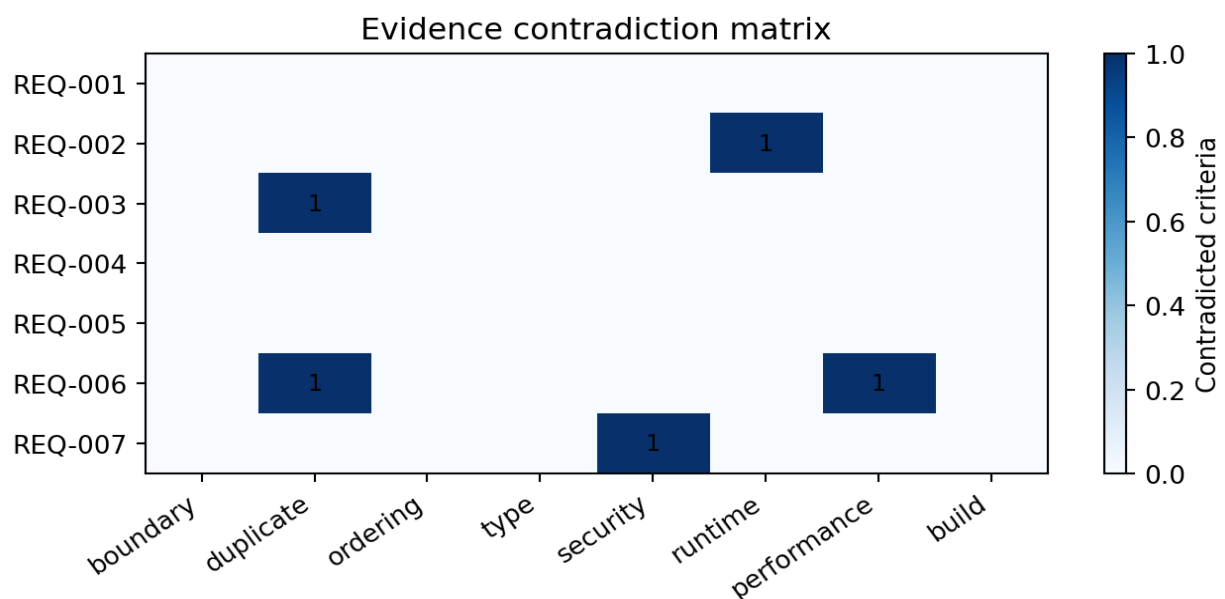
Client interpretation

This report is an assurance record, not a certification. Conclusions apply to the submitted project snapshot and the capabilities exercised in this run. Evidence-backed requirement statuses are the primary decision signal.

3. Evidence-backed verification

Important conclusions are represented as expected behavior, observed behavior, source evidence, and resulting status.

Criterion	Status	Expected	Observed / evidence
REQ-001-AC-01	VERIFIED	title=PLUMB_PROBE, status=done, priority=7	title='PLUMB_PROBE', status='done', priority=7
REQ-001-AC-02	VERIFIED	All supplied task fields preserved	Task(id=1, title='PLUMB_PROBE', status='done', priority=7)
REQ-002-AC-01	VERIFIED	Valid CSV output with required header	header=['id', 'title', 'status', 'priority']
REQ-002-AC-02	VERIFIED	Only records matching requested status are exported	rows=[[1, 'TODO-A', 'todo', '1']]
REQ-002-AC-03	FAILED	Empty filter returns header only	empty_rows=['id', 'title', 'status', 'priority'], [1, 'TODO-A', 'todo', '1']]
REQ-003-AC-01	VERIFIED	Exact duplicate occurs once in merged result	ids=[1, 2]
REQ-003-AC-02	FAILED	Distinct records sharing a title remain distinct	ids=[1, 2], titles=['ALPHA', 'SHARED'], statuses=['todo', 'todo']
REQ-003-AC-03	VERIFIED	Retained records preserve insertion order	ids=[1, 2]
REQ-006-AC-01	FAILED	No disproportionate runtime growth as input size increases	Observed 279.69x runtime growth from input 25 to 200
REQ-004-AC-01	VERIFIED	GET /health is reachable and provides the requested capability	HTTP 200 {"status": "ok"}
REQ-005-AC-01	VERIFIED	GET /items/count is reachable and provides the requested capability	HTTP 200 {"count": 99}
REQ-007-AC-01	FAILED	Do not expose hardcoded credentials or execute user-controlled commands with unsafe shell semantics.	ADMIN_API_TOKEN = 'sk_live_9f8a7b6c5d4e3f2a1b0c'



4. Testing strength

The submitted project test suite reports 4 passed and 0 failed. Mutation testing injected 1 valid code changes; 0 were caught and 1 survived. A 0% mutation score means the tests detected that share of the injected faults.



File	Function	Mutation	Outcome
server.py	do_GET	==->!=	survived

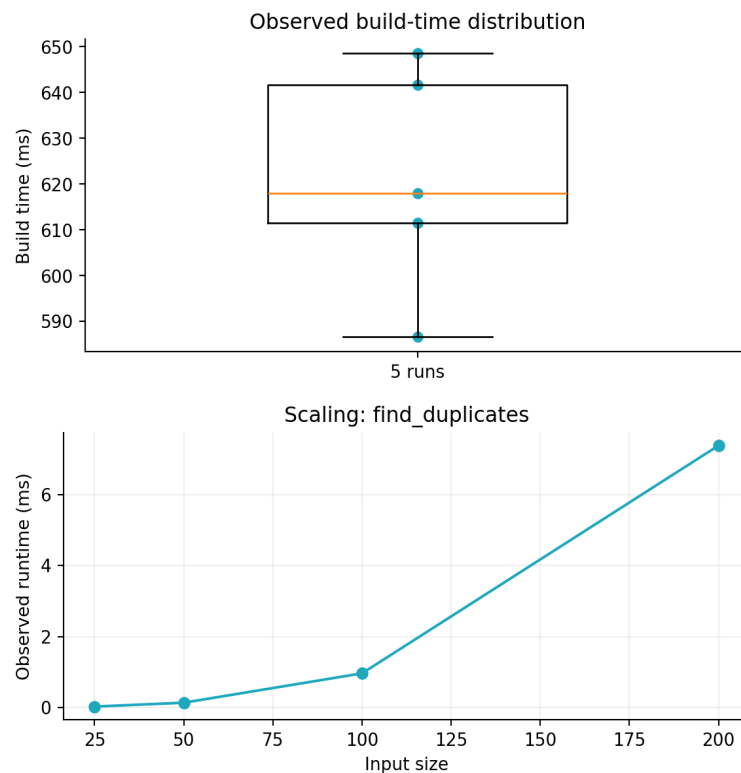
Adversarial / mutation observations

GET /health: Adaptive planner matched this requirement to the discovered route; runtime execution confirms capability existence. Expected: GET /health is reachable and provides the requested capability. Observed: HTTP 200 {"status": "ok"}.

GET /items/count: Adaptive planner matched this requirement to the discovered route; runtime execution confirms capability existence. Expected: GET /items/count is reachable and provides the requested capability. Observed: HTTP 200 {"count": 99}.

5. Performance and runtime verification

Performance measurements are separated by purpose. Timing values describe observed runs, not universal guarantees.



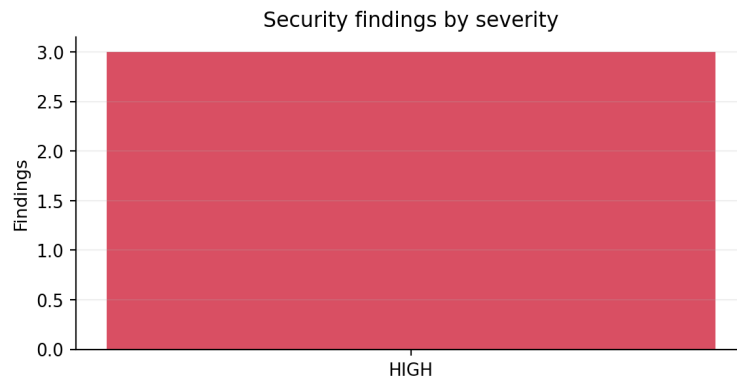
Runtime verification: EXECUTED | Target checks: 1 verified / 1 failed | Entry: server.py

Endpoint	Expected	Observed	Status
GET /health	HTTP 200 status=ok	HTTP 200 {"status": "ok"}	VERIFIED
GET /items/count	HTTP 200 count=3	HTTP 200 {"count": 99}	FAILED

Performance interpretation

Observed timings are environment-specific. A scaling failure indicates that the measured growth pattern exceeded the bounded verification expectation; it should be reproduced under production-like load before capacity commitments are made.

6. Security and architecture



Severity	Rule	File	Line	Evidence
HIGH	hardcoded-secret	admin.py	3	ADMIN_API_TOKEN = 'sk_live_9f8a7b6c5d4e3f2a1b0c'
HIGH	shell-true	admin.py	5	subprocess.run(user_command, shell=True)
HIGH	pickle-load	admin.py	7	return pickle.loads(raw_bytes)

No project-internal import relationships detected

Project inventory

File	Language	Lines	Symbols
admin.py	Python	8	2
server.py	Python	19	3
slow.py	Python	8	1
taskmgr/__init__.py	Python	0	0
taskmgr/export.py	Python	14	2
taskmgr/merge.py	Python	9	1
taskmgr/store.py	Python	14	3
tests/test_taskmgr.py	Python	16	4

7. What remains unresolved

33.3% of acceptance criteria are not fully verified in this run. These are not all the same kind of gap; PLUMB separates failed behavior from insufficient evidence.

Requirement	Criterion	Status	Criterion / reason
REQ-002	REQ-002-AC-03	FAILED	an empty filtered result must still return a valid CSV header.
REQ-003	REQ-003-AC-02	FAILED	exact duplicate tasks should be removed, but distinct tasks that share a title must remain distinct
REQ-006	REQ-006-AC-01	FAILED	Handle larger duplicate-detection workloads without disproportionate runtime growth.
REQ-007	REQ-007-AC-01	FAILED	Do not expose hardcoded credentials or execute user-controlled commands with unsafe shell semantics.

Verification capabilities

Capability	Status
static_analysis	EXECUTED
python_compile	EXECUTED
pytest	EXECUTED
runtime_http	EXECUTED
mutation_python	EXECUTED
performance	EXECUTED
container_isolation	UNSUPPORTED
ai_assisted_reasoning	UNSUPPORTED

Current execution is local-first and subprocess-based. This is not Docker-grade isolation. Unsupported capabilities are not treated as passes.

8. Final assessment

PLUMB reached **66.7%** acceptance-criteria closure and **69.5%** implementation depth. The deterministic final verdict for this verification run is **FAILED**. Test resilience is **WEAK**: none of the injected faults were detected. The summary score (57.9/100) is contextual; requirement-level evidence remains authoritative.

Interpretation

A VERIFIED status means the defined applicable evidence for that criterion passed. FAILED means the recorded verification contradicted the expected behavior. INCONCLUSIVE means the available evidence does not justify a definitive result. UNSUPPORTED means the current environment cannot perform the required check.

Evidence principle

All completion messages are claims. PLUMB uses project inspection, actual tests, observations, measurements, and recorded evidence to determine what can be stated with confidence.

Audit trail and reproducibility

The verification ID, project hash, executed probes, observed outputs, findings, and capability limitations travel with this report. The summary score is a navigation aid; requirement statuses and their cited evidence remain the authoritative conclusions.

Verification ID: 63932d95-d6c3-4c9c-85bd-d84a8bdd23b1 | Project hash: 46276a962d01b1b6... | Report generated from the same stored verification record used by the application.